

Lecture 13 - April 24 - 2026

11.9 Representation of Functions as Power Series.

General idea:

$$\underbrace{\sum_{n=0}^{\infty} C_n (x-a)^n}_{\text{power series.}}$$

$$f(x) = \sum_{n=0}^{\infty} C_n (x-a)^n \text{ we can define Dom } f.$$

If $f(x)$ is a function,
we know $\text{Dom } f$.

what is its power series?

This is important b/c, $f(x) = \frac{\sin x}{x}$; $\text{Dom } f = \mathbb{R} - \{0\}$.

$$\bullet f'(x) = \frac{d}{dx} (\sin x \cdot x^{-1}) = \frac{\cos x}{x} - \frac{\sin x}{x^2} = \frac{x \cos x - \sin x}{x^2}$$

$$\bullet \int \frac{\sin x}{x} dx = \text{DNE. i.e., there is not a function } F(x) \text{ s.t.}$$

$$F'(x) = \frac{\sin x}{x}.$$

The only way is try to approx $\frac{\sin x}{x}$ by a power series:

$$\frac{\sin x}{x} = \sum_{n=0}^{\infty} C_n x^n$$

$$\int_1^2 \frac{\sin x}{x} dx \approx \int_1^2 \left(\sum_{n=0}^{\infty} C_n x^n \right) dx$$

$$= \sum_{n=0}^{\infty} \int_1^2 C_n x^n dx$$

$$= \sum_{n=0}^{\infty} C_n \int_1^2 x^n dx$$

$$= \sum_{n=0}^{\infty} C_n \left. \frac{x^{n+1}}{n+1} \right|_1^2$$

$$= \sum_{n=0}^{\infty} C_n \left(\frac{2^{n+1}}{n+1} - \frac{1}{n+1} \right).$$

Representation of Functions using Geometric Series.

$$\frac{1}{1-x} = 1 + x + x^2 + \dots = \sum_{n=0}^{\infty} x^n \quad ; \quad |x| < 1.$$

Example: Express $1/(1+x^2)$ as a power series and find the interval of convergence.

Sln: $\frac{1}{1-y} = \sum_{n=0}^{\infty} y^n \quad ; \quad |y| < 1.$ let $y = -x^2$

$$\frac{1}{1-(-x^2)} = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

let $a_n = (-1)^n x^{2n}$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{|x|^{2n}} = |x|^2 = x^2 < 1. \quad \text{So, } x \in (-1, 1).$$

if $x = -1$; then $\sum_{n=0}^{\infty} (-1)^n (1)^{2n}$ which is divergent.

if $x = 1$ then $\sum_{n=0}^{\infty} (-1)^n (1)^{2n}$ which is divergent.

The easy way: We have $|y| < 1$ then $y = -x^2 \quad |-x^2| < 1 \Leftrightarrow x^2 < 1.$
So, $x \in (-1, 1).$

Example: Find a power series representation for $1/(x+2)$.

Sln: Using:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad ; \quad |x| < 1.$$

$$\begin{aligned} \frac{1}{2+x} &= \frac{1}{2\left(1 + \frac{x}{2}\right)} = \frac{1}{2\left(1 - \left(-\frac{x}{2}\right)\right)} = \frac{1}{2} \cdot \frac{1}{1 - \left(-\frac{x}{2}\right)} \\ &= \frac{1}{2} \cdot \sum_{n=0}^{\infty} \left(-\frac{x}{2}\right)^n = \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2^n} = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2^{n+1}}. \end{aligned}$$

$$\left| -\frac{x}{2} \right| < 1 \quad (\Leftrightarrow) \quad |x| < 2. \quad \text{Hence.}$$

$$\frac{1}{2+x} = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2^{n+1}}; \quad |x| < 2.$$

Exmpl: Find a power series representation of $\frac{x^3}{x+2}$.

Sln:
$$\frac{x^3}{x+2} = x^3 \cdot \frac{1}{x+2} = x^3 \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2^{n+1}} = \sum_{n=0}^{\infty} \frac{(-1)^n x^3 x^n}{2^{n+1}}.$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+3}}{2^{n+1}}.$$

$\text{Dom}(x^3) = \mathbb{R}. \quad \text{Dom}\left(\frac{1}{x+2}\right) = (-2, 2) \quad (\text{in the power series sense}).$

$\text{Dom}\left(\frac{x^3}{x+2}\right) = \text{Dom}(x^3) \cap \text{Dom}\left(\frac{1}{x+2}\right) = (-2, 2).$
in the power series sense.

$$\frac{x^3}{x+2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+3}}{2^{n+1}}; \quad |x| < 2.$$

Differentiation and integration of Power series.

Thm: If the power series $\sum_{n=0}^{\infty} C_n (x-a)^n$ has radius of convergence $R > 0$ then the function

$f(x) = \sum_{n=0}^{\infty} C_n (x-a)^n$ is differentiable (and therefore continuous) on the interval $(a-R, a+R)$. and:

i).
$$f'(x) = \frac{d}{dx} \left(\sum_{n=0}^{\infty} C_n (x-a)^n \right) = \sum_{n=0}^{\infty} C_n \frac{d}{dx} ((x-a)^n) = \sum_{n=1}^{\infty} C_n n \cdot (x-a)^{n-1}$$

ii).
$$\int f(x) dx = \int \left(\sum_{n=0}^{\infty} C_n (x-a)^n \right) dx = \sum_{n=0}^{\infty} C_n \int (x-a)^n dx$$

$$= \sum_{n=0}^{\infty} \frac{C_n}{n+1} (x-a)^{n+1} + C.$$

The radius of convergence of i). and ii). are both R .

Example: Express $\frac{1}{(1-x)^2}$ as a power series.

Sln:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$\frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{d}{dx} \left(\sum_{n=0}^{\infty} x^n \right)$$

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} \frac{d}{dx} (x^n) = \sum_{n=1}^{\infty} n x^{n-1} = \sum_{n=0}^{\infty} (n+1) x^n, \quad |x| < 1.$$

Example: Find a power series representation for $\ln(1+x)$ and its radius of convergence.

Sln: $\frac{d}{dx} (\ln(1+x)) = \frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty} (-1)^n x^n.$

So $\int \left(\frac{d}{dx} (\ln(1+x)) \right) dx = \int \left(\sum_{n=0}^{\infty} (-1)^n x^n \right) dx.$

$$\ln(1+x) = \sum_{n=0}^{\infty} (-1)^n \int x^n dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}, \quad |x| < 1.$$

For you! $\int$ • Find a power series representation for $f(x) = \arctan x$.
 $\{$ • Evaluate $\int \frac{1}{1+x^2} dx$ as a power series.